ApexPetro Sovereign Dataset v3 — Industrial Intelligence Report

Comprehensive Architecture, Inventory, Dimensional Metrics & Validation Audit for AI Workbench

Dataset Edition: v3.0 Industrial Intelligence
Operating Units: CDU-4, HCU-1, DISP-1, UTY
Master Entities: 112 Assets, 59 Instruments
Total Expanded Files: 204 Files
Total Storage Size: 5.14 MB (5,259 KB)
Telemetry & Event Records: 180,000+ Records
Compound Scenarios: 105 Scenarios
Mishap Benchmarks: 105 Tests
17-Point Audit: 17 / 17 Passed (100%)

1. Industrial Intelligence Domains Overview

Domain Directory	Domain Title	Files	Key Features & Scope
condition_monitoring	Condition Monitoring	8	Asset health (53.5k), vibration (33.7k), thermal (15.6k), oil analysis, abnormal register
risk_intelligence	Risk Intelligence & Anomalies	15	214 leading indicators, 524 early alerts, 105 compound scenarios, 5.2k anomaly parquet
human_factors	Human Factors Intelligence	8	2,050 fatigue records, shift cognitive workload, procedural deviations, operator actions
organizational_factors	Organizational Factors	8	1,050 safety culture indicators, management pressure, MOCs, decision delays
environment	Environmental Conditions	5	10,800 ambient records, extreme weather history, emissions and outfall monitoring
utilities	Utilities & Support Systems	8	10.5k steam, 10.5k cooling water, 10.5k electrical, utility dependency matrix
process_safety	Process Safety Intelligence	8	520 API 754 events, safety barriers, BowTie models, safety-critical equipment (SCE)
events	Event & Causal Intelligence	8	2,150 master events JSONL, 2,150 timeline CSV, 1,050 causal links, causal chain library
production_intelligence	Production Intelligence	8	Operational stability, system stress index, complexity, resilience, discipline indexes
business_context	Business Context & Indirect Risk	7	Production pressure, budget pressure, spare parts risk, vendor dependency
reliability	Reliability & Failure Prediction	7	Failure precursors with realistic windows, Weibull library, bad actors, repeat failures
integrity	Integrity & Long-Term Degradation	7	10,500 wall thickness UT records, corrosion history, API 510/570 remaining life
TEST_DATA/mishap_prediction_tests	Mishap Benchmark Suite	106	105 curated JSON test cases with evidence, leading indicators, and ground truth
tools	Validation Tooling	2	Standalone automated 17-point validation suite & expansion inventory audit

2. Detailed Dataset Dimensions (Rows, Columns, Sizes)

File Path	Format	Rows	Cols	Size (KB)
condition_monitoring/abnormal_condition_register.xlsx	XLSX	32	17	10.3
condition_monitoring/asset_health_history.parquet	PARQUET	53570	17	567.1
condition_monitoring/condition_monitoring_events.xlsx	XLSX	36	16	9.4
condition_monitoring/degradation_indicators.xlsx	XLSX	46	16	10.7
condition_monitoring/equipment_condition_assessment.xlsx	XLSX	113	18	14.5
condition_monitoring/lubrication_condition_history.xlsx	XLSX	41	17	9.3
condition_monitoring/thermal_condition_history.parquet	PARQUET	15600	15	145.7
condition_monitoring/vibration_condition_history.parquet	PARQUET	33750	16	287.6
risk_intelligence/compound_risk_scenarios.json	JSON	105	JSON	151.3
risk_intelligence/dynamic_risk_register.xlsx	XLSX	41	21	12.2
risk_intelligence/early_warning_alerts.xlsx	XLSX	525	16	55.3
risk_intelligence/emerging_risk_register.xlsx	XLSX	18	12	7.4
risk_intelligence/lagging_indicator_register.xlsx	XLSX	25	13	7.6
risk_intelligence/leading_indicator_register.xlsx	XLSX	215	15	26.5
risk_intelligence/risk_correlation_matrix.xlsx	XLSX	13	13	6.2
risk_intelligence/risk_escalation_history.xlsx	XLSX	22	13	7.6
risk_intelligence/risk_heatmap_history.xlsx	XLSX	41	11	7.6
risk_intelligence/anomaly_dataset/anomaly_event_history.parquet	PARQUET	5200	15	160.8
risk_intelligence/anomaly_dataset/anomaly_explanations.json	JSON	1	JSON	1.6
risk_intelligence/anomaly_dataset/anomaly_ground_truth.xlsx	XLSX	30	9	7.0
risk_intelligence/anomaly_dataset/anomaly_labels.xlsx	XLSX	10	7	6.2
risk_intelligence/anomaly_dataset/anomaly_patterns.json	JSON	1	JSON	1.5
risk_intelligence/anomaly_dataset/normal_behavior_patterns.json	JSON	2	JSON	1.4
human_factors/communication_quality_events.xlsx	XLSX	25	12	7.2
human_factors/fatigue_risk_indicators.xlsx	XLSX	2051	14	134.9
human_factors/human_error_precursor_register.xlsx	XLSX	18	10	6.8
human_factors/operator_action_history.xlsx	XLSX	55	13	9.1
human_factors/procedural_deviation_register.xlsx	XLSX	35	15	9.2
human_factors/shift_workload_history.xlsx	XLSX	150	12	13.1
human_factors/training_competency_risk.xlsx	XLSX	45	10	7.8
human_factors/workload_risk_analysis.xlsx	XLSX	20	10	6.4
organizational_factors/communication_effectiveness.xlsx	XLSX	51	9	7.4
organizational_factors/decision_delay_register.xlsx	XLSX	22	9	6.9
organizational_factors/management_pressure_indicators.xlsx	XLSX	100	12	10.8
organizational_factors/organizational_change_register.xlsx	XLSX	28	12	7.8
organizational_factors/organizational_risk_indicators.xlsx	XLSX	18	11	6.7
organizational_factors/safety_culture_indicators.xlsx	XLSX	1051	12	65.1
organizational_factors/staffing_capacity_history.xlsx	XLSX	193	9	12.5
organizational_factors/unresolved_issue_register.xlsx	XLSX	25	12	7.5
environment/ambient_conditions.parquet	PARQUET	10800	12	228.3
environment/environmental_monitoring_history.xlsx	XLSX	45	11	7.5
environment/environmental_risk_events.xlsx	XLSX	18	12	6.9
environment/extreme_weather_events.xlsx	XLSX	6	11	6.5
environment/weather_history.xlsx	XLSX	121	11	10.2
utilities/cooling_water_performance.parquet	PARQUET	10500	13	217.7
utilities/electrical_supply_quality.parquet	PARQUET	10500	12	151.2
utilities/instrument_air_quality_history.xlsx	XLSX	60	10	7.8
utilities/nitrogen_system_history.xlsx	XLSX	60	9	7.3
utilities/steam_system_performance.parquet	PARQUET	10500	13	156.3

utilities/utility_dependency_matrix.json	JSON	1	JSON	2.4
utilities/utility_disturbance_events.xlsx	XLSX	20	12	7.3
utilities/utility_health_history.xlsx	XLSX	90	10	8.8
process_safety/barrier_health_history.xlsx	XLSX	65	10	8.3
process_safety/bowtie_scenarios.json	JSON	1	JSON	2.7
process_safety/near_miss_precursor_register.xlsx	XLSX	30	15	8.8
process_safety/process_safety_event_history.xlsx	XLSX	521	15	48.9
process_safety/risk_barrier_degradation.xlsx	XLSX	25	10	6.7
process_safety/safety_barrier_register.xlsx	XLSX	45	13	9.5
process_safety/safety_control_effectiveness.xlsx	XLSX	25	9	6.5
process_safety/safety_critical_equipment_register.xlsx	XLSX	40	11	7.8
events/anomaly_clusters.xlsx	XLSX	18	8	6.6
events/causal_chain_library.json	JSON	1	JSON	3.6
events/causal_relationships.json	JSON	1050	JSON	342.6
events/event_correlation_matrix.xlsx	XLSX	8	8	5.5
events/event_master.jsonl	JSONL	2150	JSON	1123.9
events/event_timeline.csv	CSV	2150	7	360.7
events/precursor_events.xlsx	XLSX	45	12	8.2
events/root_cause_graph.json	JSON	2	JSON	1.8
production_intelligence/abnormality_risk_index.xlsx	XLSX	137	13	12.9
production_intelligence/maintenance_pressure_index.xlsx	XLSX	137	13	12.8
production_intelligence/operational_complexity_index.xlsx	XLSX	137	13	13.2
production_intelligence/operational_discipline_score.xlsx	XLSX	137	12	12.5
production_intelligence/operational_stability_index.xlsx	XLSX	601	17	47.4
production_intelligence/process_variability_index.xlsx	XLSX	265	14	21.8
production_intelligence/resilience_score_history.xlsx	XLSX	137	12	12.2
production_intelligence/system_stress_index.xlsx	XLSX	601	16	47.8
business_context/business_risk_indicators.xlsx	XLSX	16	9	6.2
business_context/maintenance_budget_pressure.xlsx	XLSX	11	10	5.9
business_context/production_pressure_history.xlsx	XLSX	50	11	7.9
business_context/resource_constraint_history.xlsx	XLSX	18	8	6.3
business_context/schedule_pressure_history.xlsx	XLSX	16	9	6.1
business_context/spare_part_supply_risk.xlsx	XLSX	25	11	7.3
business_context/vendor_dependency_risk.xlsx	XLSX	18	9	6.3
reliability/bad_actor_equipment_history.xlsx	XLSX	16	11	7.0
reliability/degradation_pattern_library.json	JSON	1	JSON	2.0
reliability/failure_precursor_register.xlsx	XLSX	42	14	9.7
reliability/failure_probability_estimates.xlsx	XLSX	21	10	6.3
reliability/maintenance_effectiveness_history.xlsx	XLSX	41	10	7.0
reliability/reliability_metrics_history.xlsx	XLSX	81	10	8.3
reliability/repeat_failure_analysis.xlsx	XLSX	12	9	6.1
integrity/corrosion_history.xlsx	XLSX	45	10	7.8
integrity/degradation_rate_history.xlsx	XLSX	21	9	6.2
integrity/inspection_confidence_register.xlsx	XLSX	28	10	6.7
integrity/inspection_gap_analysis.xlsx	XLSX	15	10	6.2
integrity/integrity_risk_register.xlsx	XLSX	18	9	6.8
integrity/remaining_life_history.xlsx	XLSX	28	12	7.1
integrity/wall_thickness_history.parquet	PARQUET	10500	12	123.3
TEST_DATA/mishap_prediction_tests/index.json	JSON	2	JSON	33.2
tools/validate_dataset.py	PY	N/A	N/A	13.4

3. 17-Point Industrial Intelligence Audit Verification

Check ID	Audit Requirement	Result	Verification Outcome
Check 01	Duplicate IDs	**PASSED**	All primary registers verified with 100% unique identifiers.
Check 02	No Empty Mandatory Fields	**PASSED**	Zero empty mandatory cells and zero unallowed placeholders.
Check 03	Entity Foreign Key References	**PASSED**	100% of asset references align with canonical equipment inventory.
Check 04	Parent-Child Hierarchy	**PASSED**	Asset-to-Component and Instrument-to-Asset relationships intact.
Check 05	Engineering Physical Units	**PASSED**	All telemetry units conform to standard engineering conventions.
Check 06	Engineering Range Limits	**PASSED**	All telemetry values fall within valid thermodynamic and mechanical limits.
Check 07	Chronological Continuity	**PASSED**	Event timeline is strictly monotonic and spans 2024 to 2026.
Check 08	No Future Knowledge Leakage	**PASSED**	No forward knowledge leakage found in operational registers.
Check 09	No Duplicate Telemetry	**PASSED**	Telemetry datasets contain zero duplicate timestamp/asset records.
Check 10	No Duplicate Documents	**PASSED**	All scenario, SOP, and policy documents possess unique identifiers.
Check 11	Causal Consistency	**PASSED**	1,050 causal links preserve valid arrow of time and physical mechanisms.
Check 12	Orphan Event Detection	**PASSED**	All precursor signals link to valid parent entities and units.
Check 13	Explainable Derived Scores	**PASSED**	All abstract scores contain explicit contributing factors and methods.
Check 14	Alert Traceability	**PASSED**	524 early warning alerts fully trace to underlying indicators and evidence.
Check 15	Mishap Scenarios Benchmark	**PASSED**	Mishap benchmark suite verified with 105 complete scenarios.
Check 16	Instrument Sensor Spans	**PASSED**	All readings fall within calibrated transmitter physical spans.
Check 17	Revision Consistency	**PASSED**	Dataset versioning uniform and consistent across all directories.

4. Supported Sovereign AI Workbench Capabilities

AI Reasoning Capability	Dataset Foundation & Analytical Mechanism
Abnormality Detection	Detects subtle multi-variate and contextual sensor drifts across 53.5k asset health and 33.7k vibration records.
Early Warning Detection	Tracks 214 leading indicators and 524 early alerts with 24-hour to 30-day lead times before failure threshold.
Mishap Prediction	Evaluates plant state against 105 compound risk scenarios and 105 controlled mishap benchmark test cases.
Cross-Domain Risk Reasoning	Synthesizes equipment condition with human fatigue (2,050 records), management pressure, and utility stability.
Causal Chain Reconstruction	Traverses 1,050 directed causal edges and root cause graphs to isolate initiating events from downstream knock-on effects.
Process Safety Monitoring	Audits 520 API RP 754 process safety events, safety barrier proof tests, and BowTie hazard escalation paths.